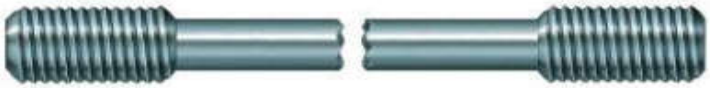
# Stress Transformation

# Stress Transformation

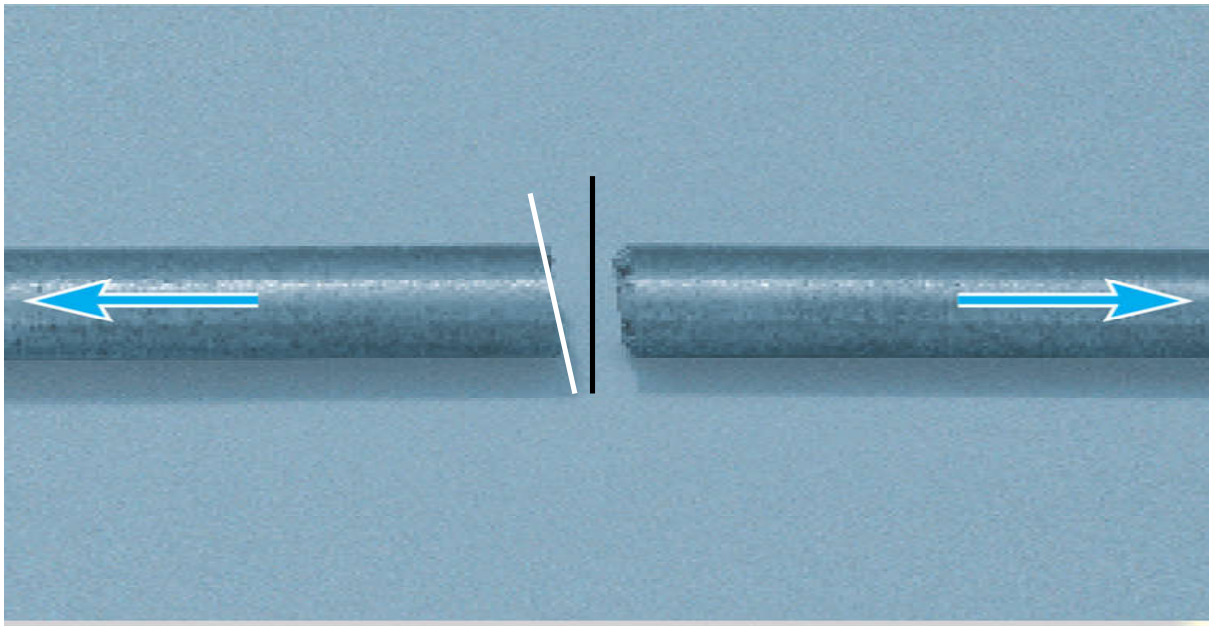
- In simple terms, the determination of the stresses in different directions.
- It is done primarily to determine the Maximum Normal Stress or Maximum Shear Stress at a point.
- Helps in predicting the direction of cracks, and the stresses that will cause them.



Fracture in a Ductile material

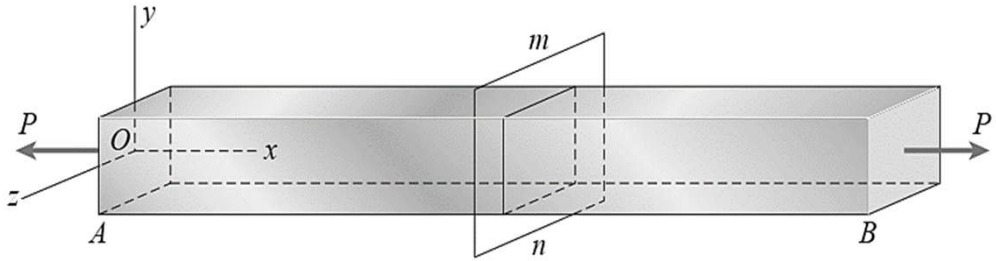


Fracture in a Brittle material

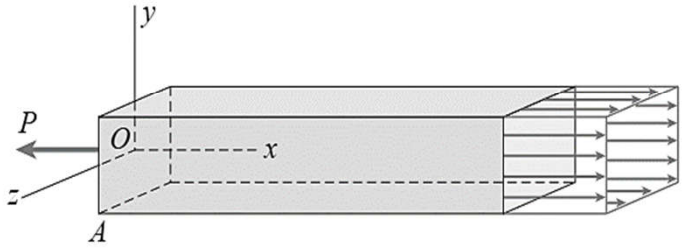


Images from Mechanics of Materials, R.C. Heibbeler

# Stress Element




(a)

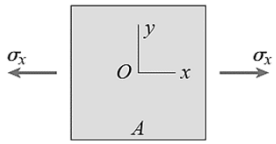



$$\sigma_x = \frac{N}{A} = \frac{P}{A}$$

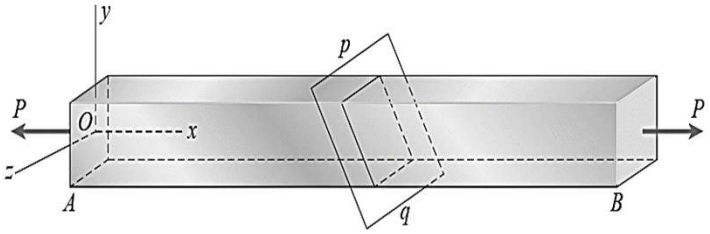
(b)



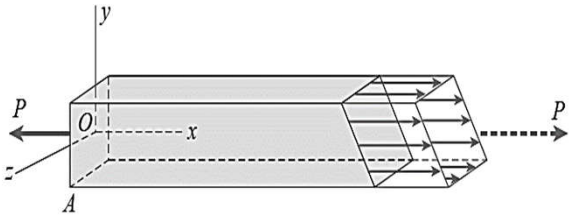

$$\sigma_x = \frac{\bar{N}}{A} = \frac{P}{A}$$

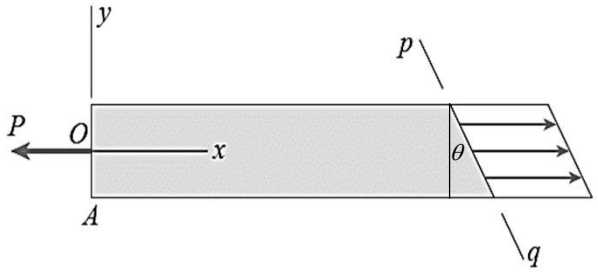


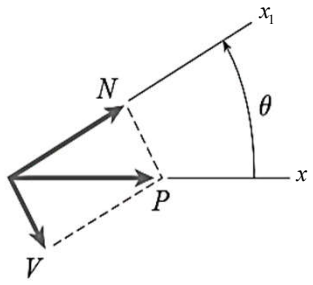
# Stress on an Inclined Plane




(a)

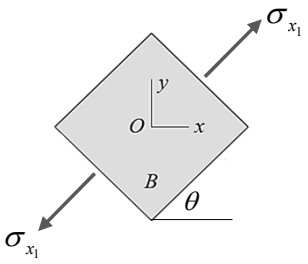


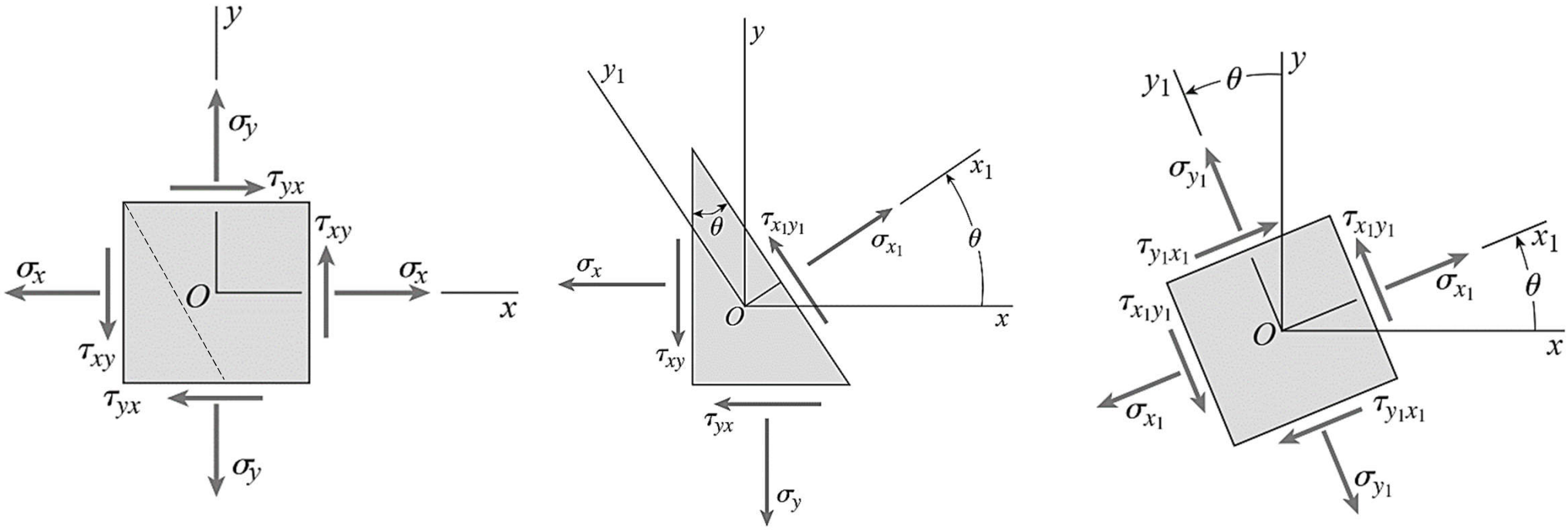

$$\sigma_{x_1} = \frac{N}{A_{pq}} = \frac{P\cos\theta}{(A/\cos\theta)}$$

$$\tau_{y_1} = -\frac{P\sin\theta}{A/\cos\theta}$$

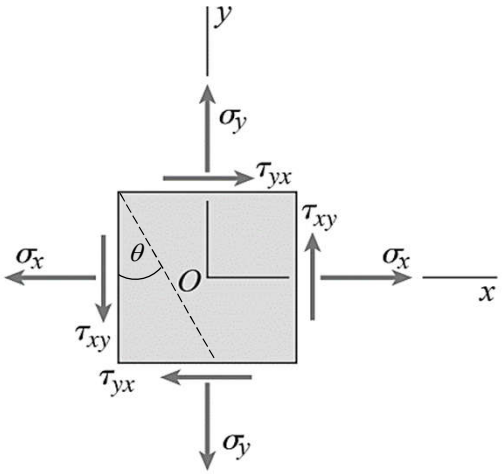


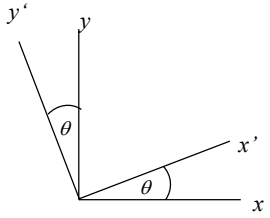
# Stress Transformation

- Basically involves the calculation of stress along rotated coordinate axes.



# Plane Stress Transformation Equations




**Sign Convention**

**Normal Stress**
Tension (+ve)
Compression (-ve)

**Shear Stress**
In the +ve direction of the axis (+)

**Angle**
+ve counterclockwise from the +ve x-axis

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(2\theta) + \tau_{xy}\sin(2\theta)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2}\cos(2\theta) - \tau_{xy}\sin(2\theta)$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin(2\theta) + \tau_{xy}\cos(2\theta)$$

$$Note: \quad \sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$$

# Principal Stresses

- When the transformed (rotated) normal stresses are their maximum and minimum possible, the are referred to as the Principal Stresses.
- They are given by;

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

- The Principal normal stresses are obtained when the stress element is rotated by $\Theta_{p.}$

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

- The Maximum Shear Stress is given by

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

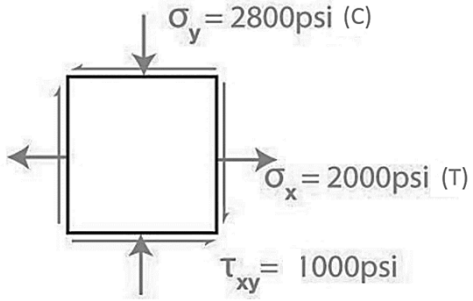
- And this occurs at

$$\tan(2\theta_s) = \frac{\sigma_x - \sigma_y}{2\tau_{xy}}$$

# Stress Transformation

Find for the stress block shown below;

1. The Principal Stresses
2. The maximum in-plane shear stress



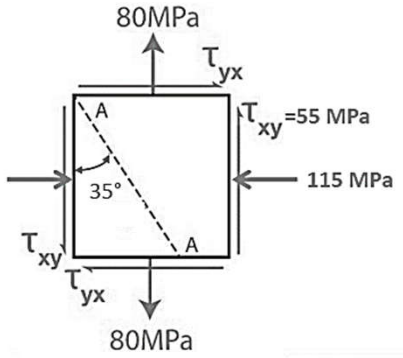

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

$$\sigma_1 = 2200 \text{ psi } (T) \qquad \sigma_2 = 3000 \text{ psi } (C)$$

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

$$\tau_{max} = 2600 \text{ psi}$$

Find the normal stress and the magnitude of the shear stress on plane A-A for the stress acting at a point in a structural member as shown.




$$\sigma_{A-A} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\theta) + \tau_{xy} \sin(2\theta)$$

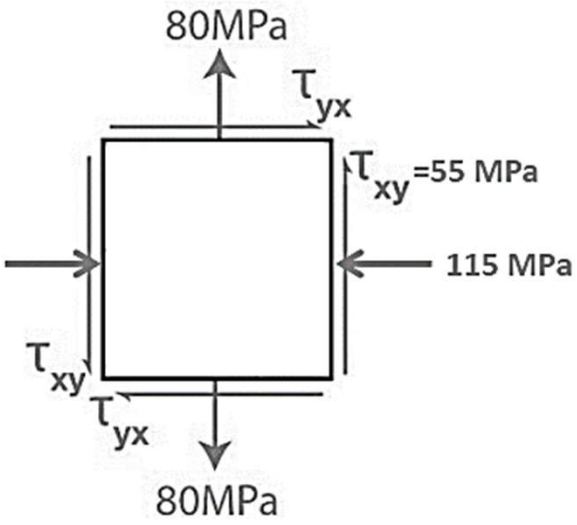
$$\tau_{A-A} = -\frac{\sigma_x - \sigma_y}{2} \sin(2\theta) + \tau_{xy} \cos(2\theta)$$

$$\sigma_{A-A} = 0.87 \text{ MPa (T)}$$

$$\tau_{A-A} = 110 \text{ MPa}$$

# Stress Transformation

Find the principal stresses and the maximum in-plane shear stress for the stress condition shown.

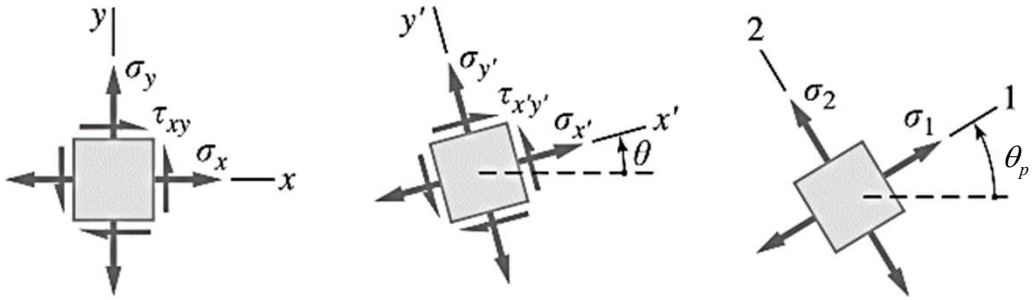


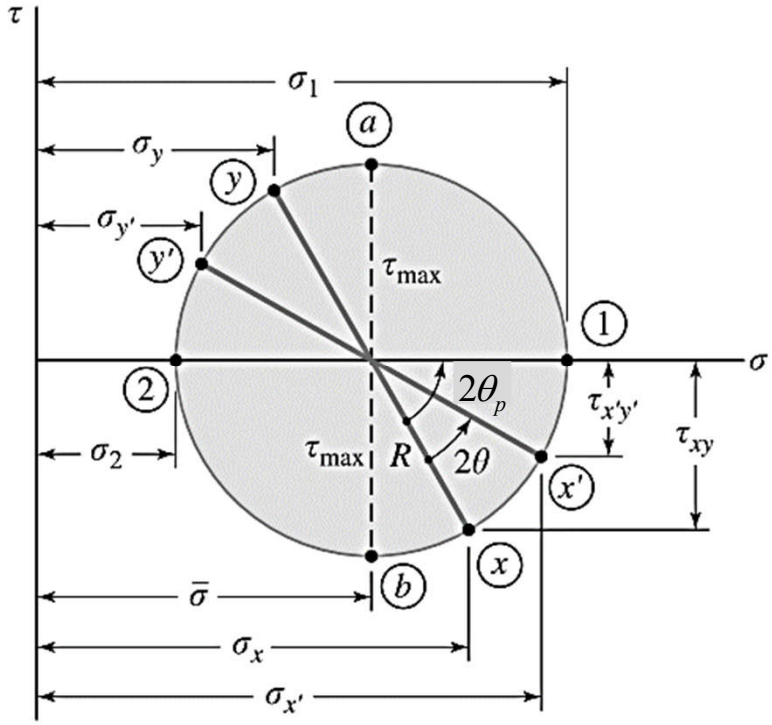
# Mohr's Stress Circle

- A graphical representation of the transformation equations.
- Upon representation of stress element data on the circle, all additional data can be obtained from the circle using trigonometry.

$$\text{Radius} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

$$\text{Centre} = \left( \frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

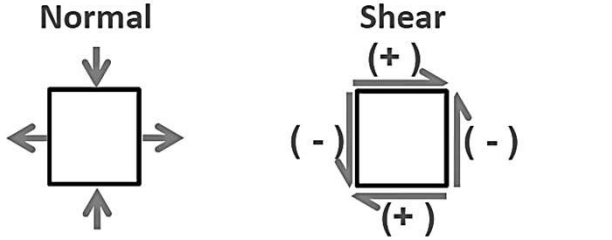


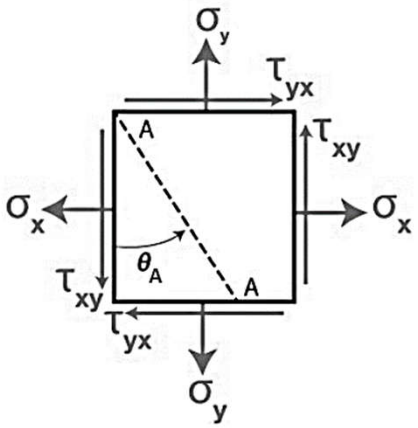




Note: angles on Mohr's circle are 2x angles on the stress element

# Sketching Mohr's circle

**Mohr's Circle Sign Convention**

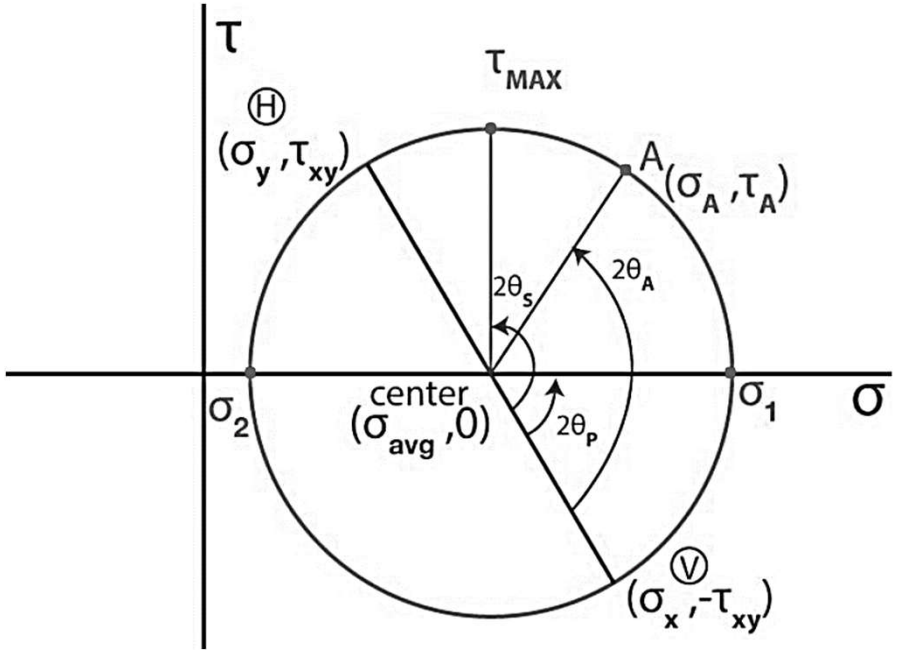



Horizontal face $H = (+\sigma_y, +\tau_{yx})$

Vertical face $V = (+\sigma_x, -\tau_{xy})$

1. Draw a set of perpendicular axes; shear stress on the vertical, and normal stress on the horizonal.
2. Read from the vertical (or $x$) face of the stress element V coordinates ($\sigma_x$, $-\tau_{xy}$), and plot them.
3. Do likewise with the horizontal (or $y$) face. i.e. H coordinates ($\sigma_x$, $\tau_{xy}$).

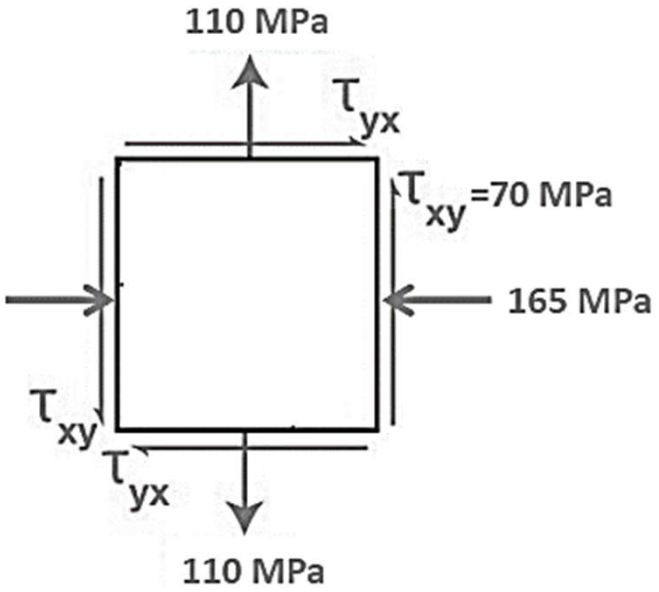


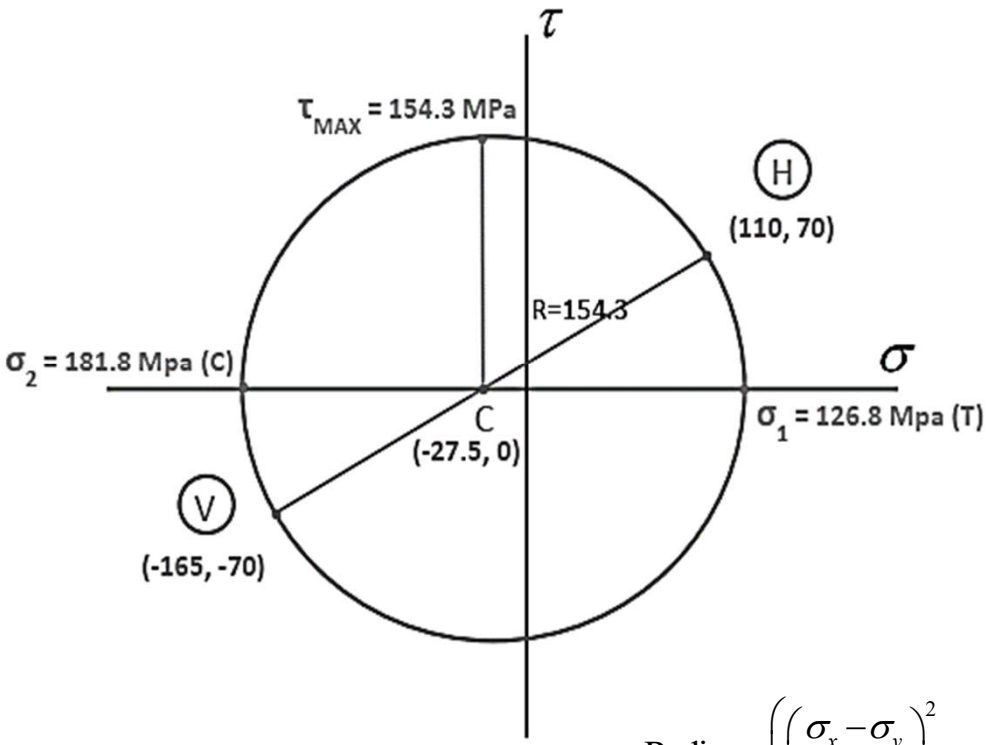

$$\text{Radius} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

$$\text{Centre} = \left( \frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

# Mohr's Stress Circle

Show a Mohr's Circle plot depicting the principal stresses and the maximum in-plane shear stress for the stress condition shown.



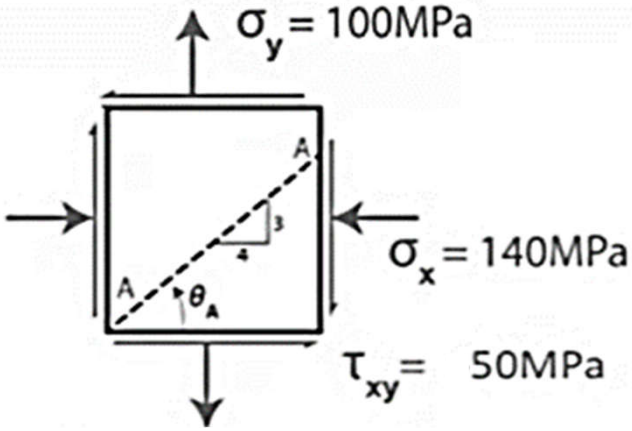

$$\text{Radius} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

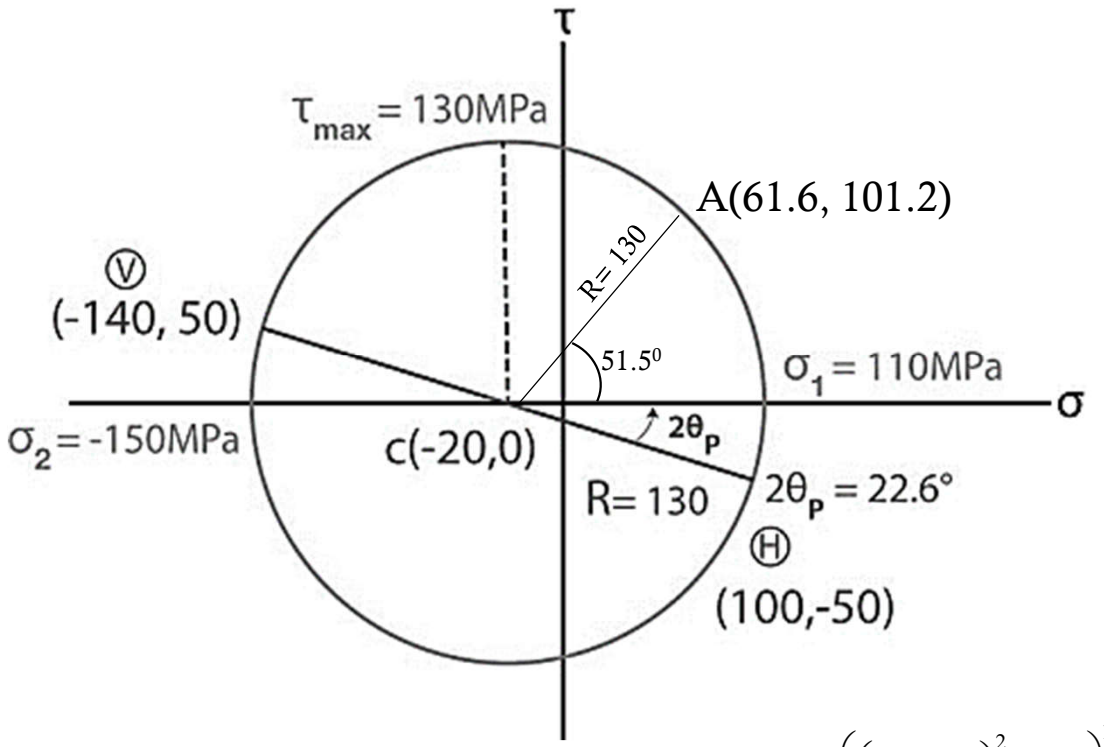
$$\text{Centre} = \left( \frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

# Mohr's Stress Circle

The stress block shown represents the stresses on two orthogonal planes through a point in a structural member. Using Mohr's circle, find;

1. The principal stresses and the maximum shear stress at that point, and show the planes on which they act on a properly oriented stress block.
2. The normal and shear stresses on plane AA through the point on a properly oriented stress block.



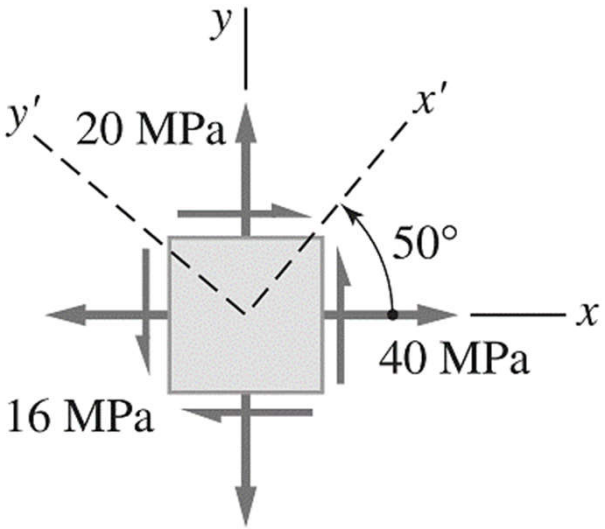

$$\text{Radius} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

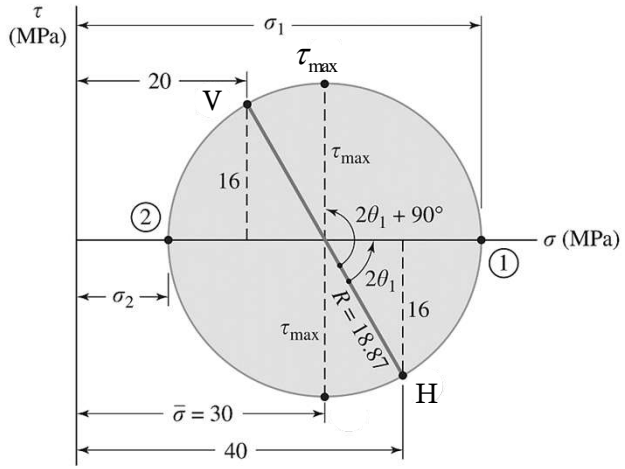
$$\text{Centre} = \left( \frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

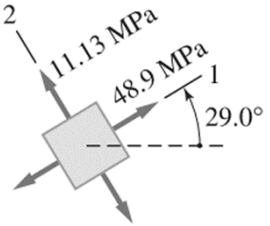
# Mohr's Stress Circle

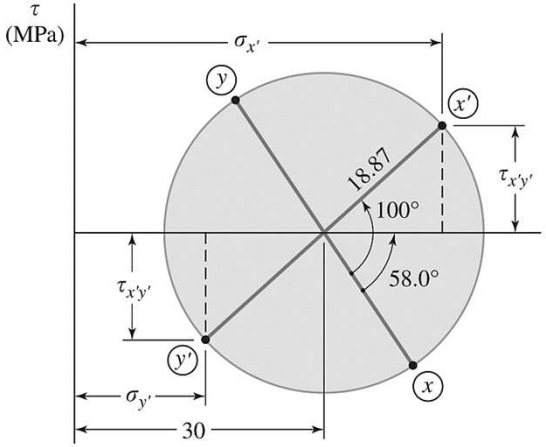
The state of plane stress a point with respect to the *xy*-axes is shown below. Using Mohr's circle, determine;

1. the principal stresses and principal planes
2. the maximum in-plane shear stress
3. the equivalent state of stress with respect to *x'y'*-axes.




$$\text{Radius} = \left( \left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^{\ 2} \right)^{1/2}$$

$$\text{Centre} = \left( \frac{\sigma_x + \sigma_y}{2}, 0 \right)$$

# Plane Strain Transformation

- Strains are customarily measured using strain gages.
- Knowing the strain in one particular direction, the strains in other directions can be determined by means of transformation equations.

**Transformation Equations**

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\theta) + \frac{\gamma_{xy}}{2}\sin(2\theta)$$

$$\varepsilon_{y'} = \frac{\varepsilon_x + \varepsilon_y}{2} - \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\theta) - \frac{\gamma_{xy}}{2}\sin(2\theta)$$

$$\gamma_{x'y'} = -\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)\sin(2\theta) + \frac{\gamma_{xy}}{2}\cos(2\theta)$$

**Principal Strains**

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \left(\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\tau_{xy}}{2}\right)^2\right)^{1/2}$$

**Principal Planes**

$$\tan(2\theta_p) = \frac{2\gamma_{xy}}{\varepsilon_x - \varepsilon_y}$$

**Maximum Shear Strain**

$$\frac{\gamma_{\max}}{2} = \frac{\varepsilon_1 - \varepsilon_2}{2} = \left(\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2\right)^{1/2}$$

## Mohr's Stress Circle

Graphical tool for depicting the transformation equations for plane stresses

**Radius =** $\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

**Center:** $\left(\frac{\sigma_x + \sigma_y}{2}, 0\right) = (\sigma_{AVG}, 0)$

The angle on Mohr's circle is 2 times the stress block angle

**Mohr's circle is a circle where each point represents the stress $\sigma$ and $\tau$ on a particular plane through a single point**

## Mohr's Strain Circle

Graphical tool for depicting the transformation equations for plane strains

**Radius =** $\sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$

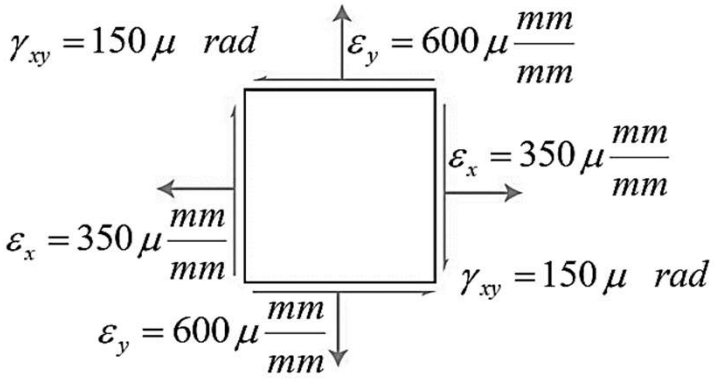
**Center:** $\left(\frac{\varepsilon_x + \varepsilon_y}{2}, 0\right) = (\varepsilon_{AVG}, 0)$

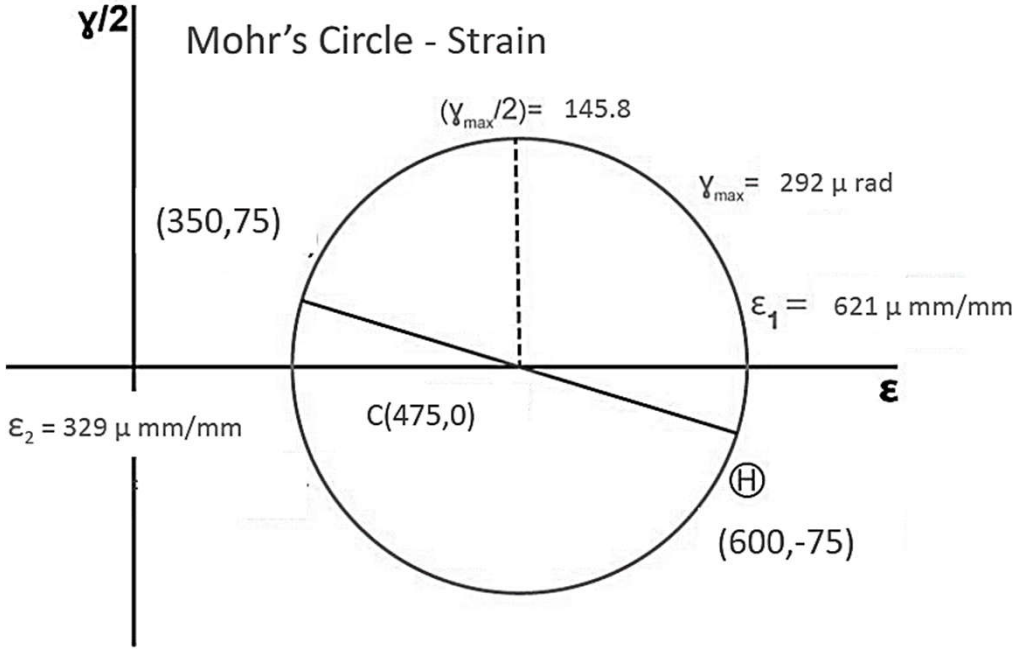
The angle on Mohr's circle is 2 times the stress block angle

**Mohr's circle is a circle where each point represents the stress $\varepsilon$ and $\gamma/2$ on a particular plane through a single point**

# Plane Strain Transformation

Using Mohr's Circle, find the principal strains and the maximum in-plane shear strain at a point, whose strain block is shown below. Also find the orientation of the principal planes from the given x-y axes.




$$\text{Radius} = \left( \left( \frac{\varepsilon_x - \varepsilon_y}{2} \right)^2 + \left( \frac{\gamma_{xy}}{2} \right)^2 \right)^{1/2}$$

$$\text{Centre} = \left( \frac{\varepsilon_x + \varepsilon_y}{2}, 0 \right)$$

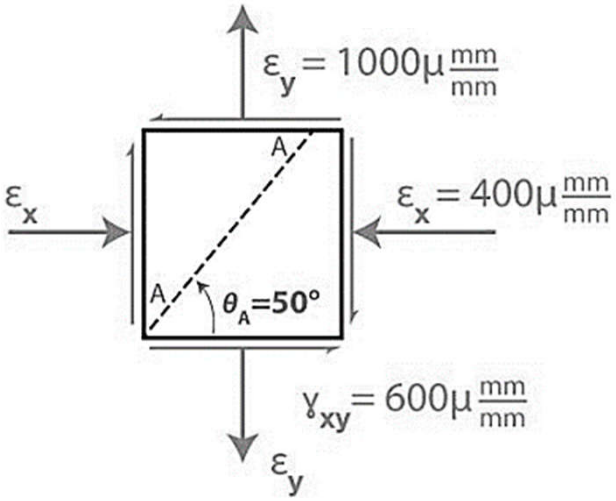
# Plane Strain Transformation

The measured strain components at a point in a body under a state of plane strain are shown.

Find:

a) The principal strains and the maximum shear strain at that point, and find the orientation of the principal planes

b) The normal and shear strains on plane AA oriented as shown.




$$\varepsilon_1 = 1016\mu$$

$$\varepsilon_2 = -416\mu$$

$$\frac{\gamma_{max}}{2} = 716\mu$$